



CHEMISTRY Department
SIMHADRI

Note No :1

Ref:SMTTPP/Chemistry/2023-24/O&M Chem/988851

Date:

12/03/2024(12:11:00)

Subject:Revision of LMI - LMI/OGN/OPS/CHEM/20 on Operating of
Condensate Polishing Units

This LMI/OGN/OPS/CHEM/20 Rev 00 "Operating Of Condensate Polishing Units" is reviewed as per the corporate OGN: OS-ISO-00- OGN/OPS/CHEM/020 Rev.No.:01 Jan2024. The following changes have been made in the LMI.

S.NO	LMI S.NO	Remarks
1	4.16	Incorporated new guidelines for failsafe operation of CONESEP type CPU regeneration system.
2	4.17	Incorporated guidelines for safe storage/ preservation of CONESEP filter.
3	4.2(l)	Anion rinse conductivity value has been revised (From 0.5 uS/cm to 0.3 uS/cm)
4	4.9	Spare resin quantity has been revised

5	4.12	Incorporated new check points in CPU periodic maintenance activities table
6	4.15	General Guidelines of CPU maintenance, points i,ii,iii added
7	4.6	C&I equipment list updated
8	4.7	Resin testing parameters changed. (parameters Total capacity, Density, Particle size distribution added)
9	7.0	Check list for records to be maintained added

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NTPC LTD

SIMHADRI SUPER THERMAL POWER PROJECT

OPERATION DIVISION FORMAL DOCUMENTATION

SYSTEM

**LOCATION MANAGEMENT INSTRUCTION-
LMI/OGN/OPS/CHEM/20, Rev. No.: 1 January 2024**

ISSUE No. : 01, Date: 11-03-2024

REVISION-01

LMI on Operating of Condensate Polishing Units



Responsibility

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LMI on Operating of Condensate Polishing Units

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1.0 INTRODUCTION

Condensate polishing unit (CPU) is an integral part of a power plant. A Condensate polishing unit in thermal power plants is used to remove dissolved contaminants such as sodium, chloride, silica, carbon dioxide etc. and suspended contaminants such as crud- iron oxide etc. from the condensate water.

Condensate polishing unit play an important role for maintaining stringent steam water chemistry parameters in super critical and ultra-supercritical units. The performance criteria of power plants equipped with condensate polishing units are that they should provide the required quality and quantity of water and be able to cope with changing operational conditions like unit startup to full load, unit flexible operation, to allow continuous operation of the unit with minor condenser seepage and allows for controlled shutdown during a condenser tube leakage.

In NTPC-Simhadri, CONESEP regeneration system is being used. It consists of three regeneration vessels: the anion regeneration/resin separation unit (ARU), Cation regeneration vessel cum mixed resins storage unit (CRU) and interface resin unit (IRU).

2.0 SUPERSEDED DOCUMENTS: OS-ISO-00-OGN/OPS/CHEM/020

Rev. No.: 0 July 2021

3.0 SCOPE : This LMI provides guidelines on operation of CPU, CPU regeneration system, good practices for resin transfer & regeneration, requirement of online monitoring instruments, management of CPU resins, periodic maintenance of CPU systems.

4.0 SPECIFIC INSTRUCTIONS TO BE FOLLOWED

Regeneration procedure of CPU

Sl.no	Activity	Responsibility
4.1	REGENERATION PROCESS : - i). Exhausted mixed resins bed is transferred from the service vessel to the ARU. - ii). After the resin is cleaned by air scrubbing, it is backwashed to remove residual corrosion product particulates and resin fines and then hydraulically separate the anion resin from the cation resin. Completion criteria for decrudding process: Effluent water turbidity < 5 NTU or Effluent water should be visually clear. - iii). After separation of mixed resins, the cation resin is transferred to CRU from bottom resin outlet line of ARU. - iv). The interface detection during resin transfer from ARU to CRU is done by online conductivity monitoring.	HOD(CHEM)

	v). During the Interface detection period when cation resin is being transferred with CO₂ injection, the conductivity value will be around 15-20 µS/cm and as soon as the cation resin is transferred and anion resin with CO₂ injection comes into contact with the conductivity the change in conductivity will indicate the detection of interface resins and transfer process will be stopped. The interface resins transferred to IRU and anion resin left in ARU. vi). Regeneration of anion and cation resins done in ARU & CRU regeneration vessels respectively. vii). After regeneration of anion and cation resin, the regenerated resin beds are rinsed with DM water to get the specified conductivity values (Ref. Sec.-4.2). viii). Then anion resin transferred to the CRU vessel and mix with air blower. After air mixing, the mixed resins bed is rinsed to the specified conductivity value (Ref. Sec.-4.2) and stored. ix). When required, this bed is used to replace an exhausted bed in one of the process vessels. x). <i>Recommendations for safe operation of cone-sep filter and safe storage of cone-sep filter mentioned in Annexure - 6</i>	
4.2	RESIN TRANSFER AND REGENERATION: Performance of CPU operation depends on the practices followed for transfer of exhaust/ fresh resins, their separation & regeneration. Importance of different steps involved, including good practices, for resin transfer & regeneration processes are described below, a) TRANSFER OF RESINS: Complete transfer of exhaust resin from service vessel to the regeneration vessel is extremely important. Exhausted resin that is left behind in the service vessel during transfer will contaminate to fresh mixed resins bed. Since this contaminated resin remains at the bottom of the service vessel it will affect the CPU effluent quality unless air scouring facility is not available at service vessel area for re-mixing. Also, resins left in the service vessel will also cause the bed to become unbalanced. b) Incomplete transfer of regenerated resins from regeneration vessel will result in short bed being put into service that can result in short run length of CPU and later lead to regeneration	HOD(CHEM)

Table-1 : Suggested DM Water flow for CONESEP Regeneration system		
Step details	Recommended flow (M³/hr)	Minimum Duration (minutes)
Fast backwash in ARU	20-30	30-60
Slow backwash in ARU	10-12	120-240
ARU-CRU resin transfer	10-12	-
ARU-IRU resin transfer	10-12	-
failure due to improper separation. c) Visual inspection of CPU service/ regeneration vessels to be done after completion of resin transfer step and any exception is to be recorded in CPU logbook. DE-CRUDING OF RESINS: As most available ion exchange capacity lies within the resins bead and not on the bead surface, it is necessary to keep the resin bead's surface free of foulants so that effective and efficient ion exchange regeneration can take place. d) Accumulation of organic or iron foulants on anion resin beads also increase bead density, making the anion resin more difficult to separate from the cation resin prior to mixed bed regeneration. COMPLETION CRITERIA: Rinse water turbidity <5 NTU or visually clear. a) Resins separation: Complete separation of resins during regeneration is most critical for the operation of condensate polishing unit. Improper separation leads to cross-contamination of the resins. Separation of resins also depends on backwash flow rate during different steps. Based on practice being followed at NTPC-Simhadri, above are the suggested flow rates (<i>Table-1</i>) for separation of resins. b) Position of flow transmitter should be after pump's recirculation line to get actual flow of water. c) Flow and pressure transmitters should be available at discharge header of DM regeneration/ resin transfer pump for monitoring and control of regeneration activities.		

	d) Resin fines removal : Open backwash of each resin lot should be done after every 10th regeneration. Open backwash of separated resins should be done in respective regeneration vessels after completion of decrudding step. e) In open backwash, carry out an extended backwash of resin beds with the top manhole cover open and observe the bed performance. The water should break through evenly. If it breaks through from one side of the vessel and floods over the bed surface, or the bed breaks into lumps, (indicating packing and channeling) backwash at the fastest possible flow rate without bringing resin over, and stir with a stout stick until the bed is level and of a consistent density. Remove any 'fines' etc. by skimming the top surface and then shoveling out. Open backwash time should be recorded in CPU logbook. f) Permanent platform should be available for inspection of regeneration vessels and open backwash of CPU resin lots in regeneration vessels. g) After completion of backwash, bed level of anion & cation resins is to be measured and top-up of resin to be done, if required. Also, records of resin top-up quantity should be recorded in CPU regeneration log sheet (Annexure-3) & CPU resins movement log sheet (Annexure-4). Regeneration of resins should be done after resins top-up to avoid any quality related issues. h) Regeneration of CPU resins: Regeneration of anion and cation resins should be done simultaneously. During anion regeneration, the temperature of sodium hydroxide solution should be maintained at 45-48 °C to improve silica elution. i) Chemical injection is to be done through envisaged program in auto mode only. Also, adequate power water flow is to be maintained during chemical injection process. j) Rinsing of resins is also an important step of CPU regeneration. Following are the recommended conductivity limits for anion, cation & mixed resins rinse processes.	
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	k) For cation rinse, effluent specific conductivity should be $\leq 3\mu\text{S}/\text{cm}$ l) For anion rinse, effluent specific conductivity should be $\leq 3\mu\text{S}/\text{cm}$ m) For mixed bed rinse, effluent specific conductivity should be $\leq 0.10\mu\text{S}/\text{cm}$	
4.3	REQUIREMENT OF RESIN MIXING IN CPU SERVICEVESSEL a) Remixing in the service vessel should always be done for 10-15 minutes after transfer of fresh resins and not to remix a bed that has already been in service since exhausted resin is likely to be put at the bottom of the bed providing a greater tendency for contaminant leakage. b) After remixing of resins, CPU must be filled and rinsed with DM water to get rinse water specific conductivity value $<01.0\mu\text{S}/\text{cm}$ and record is to be maintained.	HOD (Chemistry)
4.4	CPU SERVICE CYCLE TERMINATION CPU should be cut off from service if anyone of the following changes observed: i. In-service CPU effluent specific conductivity $>2.0\mu\text{S}/\text{cm}$ (preferably online value) ii. In-service CPU effluent cation conductivity $>0.10\mu\text{S}/\text{cm}$ preferably or CPU effluent chloride $>2\text{ppb}$ iii. In-service CPU effluent sodium $>2\text{ppb}$ iv. DP across the in-service CPU vessel (including DP of resin trap) $> 3.5\text{ kg}/\text{cm}^2$ or as specified by OEM. v. In-service CPU throughput $>$ design throughput. vi. Boiler water chloride continuously increasing and maintained above recommended limit & there is requirement of frequent CBD opening. If any one of the above changes observed in an in-service CPU, then that particular CPU should be declared exhausted and replaced with fresh mixed resins bed. However, under extreme emergency condition, if fresh mixed resins bed is not available and CPU service-in vessel outlet specific conductivity is less than $0.50\mu\text{S}/\text{cm}$, then that CPU may again be taken into service after in-situ remixing of resin bed followed by thorough DM water rinse (rinse water turbidity $<1\text{ NTU}$). However, re-mixing and service of already in-service CPU should not be adopted as a regular practice.	HOD (Chemistry)

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	resin bed can expand with decrease in flow rate allowing the corrosion products to penetrate deeper into the bed or be released in the effluent.	
	d) CPU flow changes should be made slowly. Excessive flow through the condensate polishers can cause high pressure drop and result in distortion or rupture of the under drain or screens. Total flow of service-in CPUs should be matched with actual condensate flow to check the bypass valve passing status (when 100% condensate flow is through CPUs). Any discrepancies should be recorded for corrective actions. An alarm for CPU high flow (>design flow) should be available in the CPU and unit control rooms.	HOD (Chemistry)
	e) Being sea water cooled station ,both CPU should be kept in service during start-up & during normal operation of unit and 100% condensate flow through CPU should always be ensured except CPU changeover period. CPU Cut-off parameters are same as specified in <i>Sec.4.4</i>	HOD (Chemistry))
4.6	ONLINE INSTRUMENTATIONS FOR CPU MONITORING a) For monitoring of CPU parameters, following online instruments should be made available: 1. Specific conductivity meter at outlet of each service vessel and common outlet 2. Specific conductivity analyser at outlet of each regeneration vessel(CRU/ARU/MRSV). 3. Cation conductivity meter at outlet of each service vessel and common outlet 4. Sequencer based sodium analyzer, common for one unit for measurement of sodium at outlet of each CPU. 5. Flow transmitters for power water line of acid and alkali dosing systems 6. Online concentration analysers for acid and alkali dosing systems	HOD (C&I)

	7.	Flow and pressure transmitters for DM resin transfer pump	HOD (C&I)		
	8.	Flow and pressure transmitters for regeneration air blowers			
	9.	Flow and pressure transmitters for service air blowers			
	10.	Pressure transmitter for unit side DM water common line			
	11.	DP (Differential pressure transmitter) transmitter across each service vessel			
	12.	DP transmitter for resin trap of each service vessel.			
	13.	CPU flow transmitter with integrator & reset facility for each service vessel			
	14.	Pressure transmitter for unit side resin transfer common line.			
	b) The common differential pressure limits for CPU systems recommended by most of the OEMs is given in <i>Table-2</i> .				
	Table-2: DIFFERENTIAL PRESSURE (DP) LIMITS FOR CPU SERVICE VESSELS				
	Equipment	Normal value (Kg/cm²)		High value (Kg/cm²)	Protection Provided
	CPU service vessel	≤2.1 (fresh bed), ≤ 3.5 (dirty bed, total DP across the vessel)		> 3.5	The Auto isolation of running Service Vessel by closing pneumatic inlet and outlet valves and bypass 50% & 100% flow in case of one or two service vessel isolated respectively
	Resin trap	≤ 0.5		> 0.5	Alarm only
	Common bypass	< 3.5		> 3.5	For 2x50% CPU units: Auto bypass valve (non- modulating type) will full open.

4.7	RESIN TESTING CPU resins should be tested once in a year from NETRA / OEM lab/ any reputed lab (Ref.-COS-ISO-00-OGN/OPS/CHEM/015, 'Code of Practice on Power Plant Chemistry') for Total exchange capacity, Salt splitting capacity, Water retention capacity, Total capacity, Density, Particle size distribution, any other parameters as per site requirement. RESIN SAMPLING: Resin samples should be taken from cation & anion vessels after regeneration for better interpretation of anion and cation resin analysis. A sample nozzle with ball valve should be provided from each regeneration vessels or resin outlet line of each regeneration vessels for collection of samples of anion and cation resins. Proper labeling of sample bottles should be done.	HOD (Chemistry) HOD (OFS-MM)
4.8	STORAGE OF SPARE RESINS Spare resins to be stored in moist condition in plastic drums. DM water to be used to keep resin in moist condition. The stored resins must never be allowed to become dry. Keep the resin drums under shade, preferably under shed. Stored resin water should be replaced with DM water once in a month. Spare resins can also be stored in original packing. Monthly inspection of resin bags/ drums to be done by site chemistry and record is to be maintained.	HOD (Chemistry)
4.9	GUIDELINES FOR SPARE RESINS STOCK NTPC-Simhadri, having 08 service vessels in total, one lot of Cation exchanger resin and 02 lots of Anion Exchanger resin to be kept as spare resin lots. Hydrogen form cation resin and hydroxide form anion resin to be used for CPU system. Hydrogen form Cation resin and Hydroxide form Anion resin should be procured for CPU system.	HOD (Chemistry)
4.10	MONITORING OF CPU RESINS DURING SHUT-DOWN PERIOD (Action to be taken during RSD/ long term shut down period) In an idle condition/ shut down period, each CPU service vessel must be rinsed once in a week with minimum two to three resin bed volume of water. Air mixing/scrubbing and rinsing of each CPU service vessel should also be done on fortnightly basis. A record is to be maintained. For long-term shutdown of the unit (>15 days), CPU resins should be kept in regenerated condition.	HOD (Chemistry)

4.11	IMPORTANCE OF SIGHT GLASSES IN CPU SERVICE VESSEL It is critical to remove all of the exhausted resin before a freshly regenerated resin bed is transferred into the vessel. If exhausted resin is left at the bottom of the service vessel, unexpected contaminant leakage may occur, especially if ammonium form operation is employed. Sight glass should be available in regeneration, CPU service vessels and resin transfer lines for monitoring purpose.	HOD (TMD)/HOD(Chem)	
4.12	PERIODIC MAINTENANCE ACTIVITIES OF CPU SYTSEM		
Table-5: Equipment Inspection Schedule			
Sl. No	Equipment	Check/Work to be done	Frequency
i.	Caustic storage & metering tanks	One by one tanks should be checked for the corrosion and proper condition of rubber lining (hardness and spark test), gauge glass cleaning. Inspection of fumes absorber and CO ₂ absorbers. Vessel and lines painting, joint bolt corrosion and replacement. Tank inlet & outlet valves also to be checked.	Once/2-years
ii.	Acid storage & metering tanks		Once/year Resp: OFS MM/CHEM
iii.	CPU regeneration system	P&I checking of CPU regeneration system including regeneration & resin transfer program sequences.	Once/year
iv.	CPU regeneration vessels	Inspection, checking of rubber lining of vessels, rubber hardness, acid/alkali dosing internals, tightness of strainers, cleaning of view glasses, position calibration of control valves, calibration of flow and pressure transmitters etc.	Once/2-years
v.		Inspection of Driplex ARU/CRU siphon break line and NRV (if provided), gasket of resin outlet lines (inside vessel).	Resp: OFS MM/CHEM
vi.	Cation regeneration unit	Inspection of acid injection pneumatic valve connected to cation regeneration unit.	Once/six-months Resp: C&I/CHEM

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	Note: a. During internal inspection of CPU service vessel/resin trap, its manual inlet and outlet isolating valves should be kept closed. b. During long shutdown period, CPU common inlet and outlet manual valves also to be kept closed. These valves to be opened after proper venting & re-pressurization of completed condensate pipelines. c. Pressure and leakage test of all CPU service vessels to be done at the earliest opportunity after normalization of CEP pumps post long shutdown/ overhaul of unit. d. Please refer 'OGN-OPS-SYST-018_Quality Management in Maintenance and Overhauling' for class of check. e. Responsibility for P&I check: i. For CPU regeneration system: Chemistry and C&I ii. For CPU service vessel system: Chemistry, Operation and C&I f. A detailed joint protocol is to be maintained for all activities.	
4.13	SPECIAL INSTRUCTIONS FOR CPU REGENERATIONSYSTEM 1) All regeneration process like resin de-cruding, separation, regeneration, exhaust, and fresh resin transfer etc. should be done in auto mode from the envisaged programs. 2) Already regenerated mixed resin must be rinsed to conductivity <0.10 uS/cm at regenerated area before transferring to CPU service vessel. 3) Before start of exhaust resin transfer, ensure that the CPU service vessel (selected under resin transfer) is out of service, isolated (both manual and pneumatic inlet and outlet valves should be fully closed), and depressurized. 4) After completion of exhaust resin transfer process, mixed resin bed level, colour and other details to be regeneration log sheet (Annexure-2)	HOD(Chemistry)

	5) Resin top-up quantity should be recorded in CPU regeneration log sheet (Annexure-2) & CPU resins movement log sheet (Annexure-4). 6) CPU resin lot movement log sheet (Annexure-4) must be maintained for performance monitoring of individual resin lot. 7) After resin transfer from service vessel to regeneration vessel, loose crud from resins to be removed by immediate backwash . 8) Whenever any blower is stopped, always keep the respective bleed valve in open mode. Also, air used for mixing and transfer of resin should be oil free. 9) If resins separation is disturbed by loss of flow or sudden change of flow, must perform additional separation. 10) Hot water should always be used for regeneration of anion resin. 11) After completion of acid and caustic injections, a slow rinse of anion and cation regeneration vessels is to be done for a minimum of one hour. 12) After completion of the cation resin regeneration process, backwash the cation regeneration vessel for a minimum of 30 minutes and keep the bed settling down. Check the top surface of the cation resin bed. If a layer of anion resin is seen, then this may be due to either incomplete fresh resin transfer from the regeneration vessel to the CPU vessel or to an issue in resin transfer from the ARU to the CRU after separation. 13) Recording rinse times provides an early identification of resin condition since extended rinse times indicated resin fouling. 14) Extra care should be taken for regeneration of CPU resins exposed with high level of sodium, chloride, silica etc. due to unit operation under condenser seepage/ leakage condition. a. Anion resin first regenerate with 5% caustic solution and then soaks with 6-7% caustic solution for 3-4 hours. Double quantity of caustic should be used for regeneration. b. Cation resin should be regenerated with 5-6% hydrochloric acid and soaks for 1-2 hours. Double quantity of acid should be used for regeneration. 15) Pressure gauges should be available in regeneration vessels for	HOD(Chemistry)
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	v. CPU service vessel should be rinsed properly before taking it in to service. vi. Before taking CPU in service, it should be pressurized by its dedicated pressurization line only. Direct pressurization of the CPU from main inlet line should not be done vii. CPU service vessel pneumatic valves should be operated from remote/ GIU only. viii. CPU pneumatic bypass valves should always be kept in auto mode during unit normal operation. ix. CPU service inlet & outlet pneumatic valves are also to be kept in auto mode during unit start-up and normal operation.	HOD (Chemistry)
	x. Flow totalizers with reset facility should be available for each CPU service vessels. xi. To avoid exhaustion of all CPU vessels at a time, always maintain wide difference in loading of CPUs.	HOD (C&I) HOD (Chemistry)
	xii. During unit startup period, deaerator makeup flow rate should not change very frequently to avoid any performance related issue in CPU.	HOD (Operation)
	xiii. Differential pressure (DP) is a critical parameter for CPU system, so monitoring and recording of DP across running service vessel, resin trap and common header must be ensured. Also, it is not advisable to bypass any protection related to high DP.	HOD (Chemistry)/ HOD (Operation)
	xiv. During condenser seepage/ leakage condition, ensure that both bypass valves (pneumatic & manual) of CPU must be in full closed condition, even during changeover of CPU service vessel, to avoid ingress of any contaminated condensate water in feed water circuit.	HOD (Chemistry)
	xv. CPU service vessels display & controls should be maintained available in OWS of CPU & Main plant control rooms.	HOD (C&I)/ HOD(Chemistry)

	xvi. Online conductivity and sodium analysers of CPU service vessel should be available. xvii. Specific & Cation conductivity analysers of regeneration and service vessels should be cross checked by portable conductivity & ACC analysers, respectively. Frequency: Once/week. xviii. PI tags should be available for CPU flow, conductivity, cation conductivity and sodium analysers of each CPU service vessels, pre-filter flow & DP	HOD (C&I)
	xix. Service-in CPU vessels water quality should be checked by ion chromatography instrument for reliable and accurate determination of Sodium and Chloride.	HOD (Chemistry)
	xx. After tripping of unit, CPU service vessels should be taken out of service only when there is no flow from hot well to deaerator.	HOD (Chemistry)
4.15	GENERAL GUIDELINES FOR MANAGEMENT OF CPUSYSTEM i. All maintenance jobs related to CPU service vessel area will be looked after by site turbine maintenance department. ii. All the maintenance jobs related to CPU regeneration system including resin transfer system and pipelines are to be looked after by the Site offsite maintenance department iii. CPU regeneration and operation should be monitored by the chemist, looking after the main plant chemistry area. iv. The CPU regeneration and service vessel area activity logbooks are to be reviewed by the shift chemists at least once per shift. v. At least one number spare CONESEP filter per regeneration system should be available at station for ready changeover in case of any damage in existing CONESEP filter.	HOD (OFFSITE)

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	with DM water resin flushing valves, resin inlet/ outlet valve of other CPUs of same unit etc. 5. Close feedback of CPU service inlet and outlet valves is required for operation of its vent, DM water top & bottom, resin inlet/ outlet, air inlet, nozzle flushing, and drain valves. 6. Auto close of CPU vessel inlet and outlet pneumatic valves on all CEPs OFF. 7. DM water rinse of CRU/MRSV mixed resin to conductivity <0.10 uS/cm (interlock) before transferring to CPU service vessel. 8. The anion and cation regeneration programs should be disabled during fresh resin transfer from CRU/MRSV to CPU service vessel. 9. Protection-interlock bypass register is to be maintained for CPU regeneration and service vessel systems.	
	o. Spare resin trap and inlet outlet valves for CPU service vessels should be available.	HOD (TMD)
	p. Bulk acid and caustic storage tanks of CPU regeneration system should have minimum one-month storage capacity for AVT-O operation of all associated units and considering frequent start-up/ shut-down of units due to RSD (reserve shutdown)/ maintenance work.	HOD (CHEMISTRY)

4.16	INSTRUCTIONS FOR SAFE OPERATION OF CONESEP FILTER	
1	i. The ARU vessel is provided for cleaning of resin separation of mixed resins and thereafter regeneration of anion resin. The ARU vessel is not meant for storage of exhausted resin with crud. ii. The ARU lining has been specially designed with pores. Due to storage of exhausted resin with crud in ARU, the pores may get choked. If the same is stored for overnight or longer, it will affect the lining due to corrosive nature of crud & type of crud generated.	HOD (Chemistry)
2	i. The cleaning activities (ABRO) of resins in ARU should be done in a continuous process, not intermittently. ii. After transfer of exhaust resins to ARU, immediate backwash of ARU is to be done to remove loose crud and fines from the exhaust resin bed. Keep ARU vent & backwash outlet valves open during this backwash process. This process is to be continued until the backwash outlet water is cleared. iii. Backwash flow is to be slowly increased to 20-25 M3/hr. During this process, backwash water flow and pressure to be monitored closely. Backwash water pressure at ARU CONESEP inlet (pressure gauge should be provided at downstream of ARU backwash control valve, if not available) is to be maintained <3.0 kg/cm². iv. PRV (pressure relief valve) of DM resin transfer/regeneration pump should be functional to protect the CONESEP filter from high pressure	
	vi. During air mixing, air blower pressure and flow should be monitored & recorded in CPU logbook. Tripping of air	

	blower at high discharge pressure (>0.50 kg/cm ²) should be ensured.	HOD (Chemistry)
3	i. During mixed resin backwash/ separation, ARU bottom water flow should increase slowly by backwash control valve. Healthiness of ARU backwash control valve should be ensured. ii. Air mixing time should be maintained 3 minutes for ABRO (Air bump & rinse operation) process. iii. During resin de-cruding process, changeover of air mixing blower must be done after every 5th ABRO cycle to avoid chances of damage in the bottom lining of CONESEP filter by excess heat of air. iv. During ABRO process, close monitoring of water level in the ARU vessel is required. The water level tends to increase or decrease with ABRO cycles v. If ARU vessel is drained, it should be filled slowly (flow <10 M³/hr) from backwash inlet control valve. vi. Air scrubbing (air mixing) should be done when water level is adequate (150-200mm above resin bed) in regeneration vessel vii. Air scrubbing should not be done in empty vessel. viii. Air and bottom water should not be used together in ARU/CRU regeneration vessels for scrubbing/ de-cruding of resins. ix. Few cycles of air scrubbing and rinsing of exhaust resins in CPU service vessel, will drastically reduce the de-cruding time of exhaust resins in ARU. However, minimum water level (150-200mm above resin bed in CPU service vessel should be ensured before start of air scrubbing process. Resin trap flushing should be done after completion of fresh resin transfer & rinse activities to avoid choking with resin fines/loose rust. x. Regeneration process including resin de-cruding process is to be done in auto mode through envisaged program	

	only. xi. Open backwash of each resin lot should be done after every 10th regeneration/ one year operation (whichever is earlier) to remove resin fines. Record is to be maintained. xii. Acid should not be used for cleaning of ARU CONESEP filter. xiii. Operation of regeneration vessel at high pressure (above normal working pressure) should not be done. xiv. Visual inspection of CONESEP filter should be done from top/side manhole, at least Once/six months. Inspection observations report should be maintained. 4.17 SAFE STORAGE / PRESERVATION OF CONESEP FILTER 1. Every care to be taken during transit of CONESEP filter. 2. During loading / unloading and shifting of CONESEP filter, care to be taken to avoid any jerk. 3. CONESEP filter to be stored in covered shed. 4. CONESEP filter to be stored away from oil & grease. 5. CONESEP filter to be stored away from fire items. 6. No heavy material to be stored on the CONESEP filter. 7. Preservation status of spare CONESEP filter is to be monitored and recorded. Frequency: Once/month.	HOD (Chemistry)
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5.0 REFERENCE:

OS-ISO-00-OGN/OPS/CHEM/020 Rev.No.:1 Jan2024

6.0 REVIEW

The Head of Chemistry will be responsible for reviewing this document on a 2-yearly basis or as necessary.

7.0 CHECKLIST

SI No.	Record Description	Maintained by	Frequency	Remarks
1	CPU regeneration data as per Annexure-2	Chemistry	For each regeneration	
2	CPU operation data as per Annexure-3	Chemistry	Twice/shift	
3	CPU resin movement records as per Annexure-4	Chemistry	For each resintransfer & regeneration	
4	CPU service record as per Annexure-5	Chemistry	For each service	
5	CPU resin stock and preservation details	Chemistry	Once/month	
6	CPU resins analysis.	Chemistry	Once/year	
7	Offline testing of vessel sp. Cond. and ACCanalysers	Chemistry	Once/week	
8	Offline analysis of sodium and chloride of service-in CPUs	Chemistry	Once/week	
9	Open backwash record of CPU resin lot as per Annexure-4	Chemistry	After 10 th regeneration of each resin lot	
10	Lot wise resin top up record	Chemistry	As per requirement	
11	Inspection protocols	Chemistry	As specified in Sec.-11.0	
12	P&I checking of CPU regeneration system	Chemistry	Once/year	
13	Sequence checking of regeneration, resintransfer and CPU service-in/out programs.	Chemistry	Once/year	
14	P&I checking of CPU service vessels	Chemistry	Unit OH or Once/2-years	
15	Preservation status of spare CONESEP filter	OFS	Once/month	
16	Stock and storage status of CPU spare resins	Chemistry	Once/month	

TROUBLESHOOTING FOR CONDENSATE POLISHING UNITS**(Annexure-1)**

S I N O	PROBLEMS	SUGGESTED ACTIONS
1	Regenerated Anion or Cation resin will not rinsedown	• Check rinse water conductivity by portable conductivity meter. If rinse water conductivity lab value is same, repeat air scrub and rinse again. • If still not rinsing down, collect rinse water sample for analysis. • Resin can be cross-contaminated or wrong resin added for topping off. • Resin may be fouled or cation resin may be de-sulfonated.
2	CPU showing poor performance in absence of condenser leak or other in-service problems.	• Check polisher regeneration including quality of regenerants, mixed resins separation process (cross resin or regenerant contamination), poor removal of corrosion product during resin decrudding, resin levels, sufficient rinse of individual bed before mixing, possibility of separation during transfer, proper mixing in CPU service vessel. Resins quality should be analysed.
3	In-service CPU Effluent cation conductivity and/ or Sodium/ /Chloride increasing	• Check if cation column is exhausted. • Check for condenser leak or contaminated make-up (any increase in Condensate cation conductivity value) • Determine if CPU approaching ammonia cycle. • Determine if high airin-leakage. • Check flow and pressure differential across the bed and observe sight glass to see resin surface (Uneven flow through bed/ channeling inbed) • Increase in condensate temperature • Investigate to determine if exhausted resin left in service vessel prior to transfer of regenerated bed. • Verify quality of treatment chemical. • Check for cross contamination during next regeneration.
4	Short run length	• Verify bed level appropriate (bed not short of resin). • Verify no channeling in CPU bed. • Verify adequate regeneration. • Check condensate temperature (has impact on effluent silica). • Check air in-leakage. • Determine if vessel was in service during a condenser leak or unit start-up or taken in service after a long shutdown. • Verify flow rate through bed to confirm short run length.

5	High differential pressure / low flow through bed	• Check root valves of DP transmitters • Verify bed level appropriate (bed not long or short of resin) • Verify no channeling in CPU bed. • Review regeneration log to determine adequate crud removal during regeneration. • Determine corrosion product (crud) loading in CPU inlet sample. • Determine if vessel was in service during a condenser leak or unit start-up or taken in service after a long shutdown. • Check resin trap and do physical inspection. • Look for mechanical problems, ruptured or distorted screens, junk from outage etc.
6	Resin beads in CPU vessel resin trap	• Inspect tightness of bottom strainers and nozzles of CPU vessels. Nozzles provided by only Driplex for flow of DM water from bottom during exhaust resin transfer process. • For Driplex CPU, check tightness of bottom water flange from bottom manhole of CPU. Any gap in between MSRL & SS flange may be tightened. • Open backwash of resin lot to remove resin fines.
7	Incomplete resin transfer from CPU service vessel	• Ensure that bottom water nozzles are not choked. • Check and correct orientation of bottom water nozzles during inspection of service vessel. • Ensure changeover of DM water flow through top and bottom of CPU during exhaust resin transfer. • Maintain specified DM water flow throughout the resin transfer process. • For few minutes, DM water to be taken from CPU bottom water line connected to CPU outlet header during exhaust resin transfer step. This step should to be applied when resin beads not seen in view glass of resin transfer line. • Periodic inspection of CPU service vessels should be done.

ANNEXURE-2**TEMPLATE FOR CPU REGENERATION LOG SHEET**

(For CONESEP type Regneration System)

Station Name:

Financial Year:

Month:

Exhaust resin received date		Received from vessel		Regeneration no. (current FY)		
Resin Lot No.		Cycle throughput (M3)		Exsht. Resin color	Light Brown/ Red-Brown/ Gary-black	
No. of unit startup during srvice cycle		Resin Level	Mixed resins: Low / Normal/ High Cation resin level: Low / Normal/ High Anion resin level: Low / Normal/ High			
		Resin top up	Cation (Ltr.)		Anion (Ltr.)	
Date	PROCESS DETAILS		DM regn/ resin transfer pump flow (M3/hr)	Pump totalizer readings (M ³)		DM Water consumed (M ³)
	1. RESINS TRANSFER			Initial	Final	
	1-a. Exhaust resin transfer to ARU					
	1-b. Vessel cleaning					
	1-c. Fresh resin transfer & rinse					
	Total Water used for resins transfer (A)					
	2. REGENERATION PROCESS					
	2-a. Exhaust resin decrudding (backwash/ ABRO)					
	Total no. of ABRO cycle:					
	2-b. Mixed resin separation & transfer					
	Water used in resins decrudding & separation processes (B)					
	3-a. Acid-Caustic injection & displacement: Concen.(%): HCl....., NaOH.....temp (°C)..... Chemicals used (MT): HCl (30%)..... NaOH (100%).....					
	3-b. CRU cleaning & final rinse					
	3-c. ARU classification & final rinse					
	3-d. ARU & CRU final rinse					
	3-e. ARU to CRU transfer, mixing & final rinse					
	3-f. Mixed resins rinse before transfer to service vessel					
	Water used for regeneration of resins (C)					
	4. MISCELLANEOUS ACTIVITIES					
	4-a. ARU vessel backwash (if required)					
	4-b. CRU vessel backwash (if required)					
	4-c. CPU vessel rinse & service (Lot no:)					
	4-d. CPU vessel rinse & service (Lot no:)					
	Water used for miscellaneous activities (D)					
Fresh resins transfer date:			Transfer to CPU service vessel:			
REMARKS: 1. Total number of ABRO cycles: 2. Total DM water consumed for resins transfer & regeneration (A+B+C), M ³ : 3. Any issues faced during regeneration & tresnfer: 4. Time consumed for rinse of cation & anion resins after their regeneration to achieved rinse water specific conductivity 3.0 & 5.0 µS/cm respectively: Cation resin:.....minutes, Anion resin:.....minutes						

ANNEXURE-3**TEMPLATE FOR CPU PERFORMANCE LOG SHEET****PROJECT:****STAGE:****UNIT:****MONTH:**

Date	Time (hrs)	Unit Load (MW)	CPU Inlet Temp	CPU bypass v/v position (%open)	CPU common DP (Kg/cm2)	CPU Serv ice Vessel Flow (M ³ /hr)			CPU Totalize r flow (M ³)			CPU Vessel DP (Kg/cm ²)			CPU Res in Tra p DP (Kg/cm ²)			CPU Vessel Outlet									Silica ppb Lab
						A	B	C	A	B	C	A	B	C	A	B	C	Sp Cond. (μs/cm)			ACC (μs/cm)			Sodium (ppb)			
																		A	B	C	A	B	C	A	B	C	

ANNEXURE-4
TEMPLATE FOR CPU RESIN MOVEMENT LOG
SHEET

Sl. No		Resin Lot-1	Resin Lot-2	Resin Lot-3	Resin Lot-4	Resin Lot-5	Resin Lot-6	Resin Lot...
1	Date of transfer to regeneration unit							
2	Resin lot open backwash done (Yes/ No)							
3	Resin top up quantity (Cation/ Anion) in ltr							
4	Date of Regeneration (<i>date of chemicals injection</i>)							
5	Mixed resins Levels after regeneration (Normal/ Low/ High)							
6	Date of transfer to CPU service vessel							
7	Transfer-in Service Vessel (A/B/C/D)							
8	CPU In-service date							
9	CPU in-service parameters (Sp. cond/ ACC/ Na/ SiO ₂)							
10	CPU Service-out date							
11	Service throughput (M ³)							
12	CPU service-out parameters (Sp. cond/ ACC/ Na/ SiO ₂)							
13	Number of unit startups experienced							
Remarks:								
1	Date of transfer to regeneration unit							
2	Resin lot open backwash done (Yes/ No)							
3	Resin top up quantity (Cation/ Anion) in ltr							
4	Date of Regeneration (<i>date of chemicals injection</i>)							
5	Mixed resins Levels after regeneration (Normal/ Low/ High)							
6	Date of transfer to CPU service vessel							
7	Transfer-in Service Vessel (A/B/C/D)							
8	CPU In-service date							
9	CPU in-service parameters (Sp. cond/ ACC/ Na/ SiO ₂)							
10	CPU Service-out date							
11	Service throughput (M ³)							
12	CPU service-out parameters (Sp. cond/ ACC/ Na/ SiO ₂)							
13	Number of unit startups experienced							
Remarks:								

ANNEXURE-5
SOP FOR CPU SERVICE DURING UNIT START UP
& NORMAL OPERATION

SI No	Sequence of Operation for taking CPU in-service	Action/ Status
1	CPU filling	Initial filling of CPU will be done by opening of selected CPU vent valve and DM water Inlet valve and running DM regeneration/ resin transfer pump.
2	CPU rinsing & pressurization with DM water pump	CPU rinsing to be done with DM regeneration/ resin transfer pump to get CPU outlet sp. cond. <0.10 $\mu\text{S}/\text{cm}$. Pressurize CPU vessel by DM regeneration/ resin transfer pump. Note: i. If sample pressure is low then portable conductivity meter (low level detection) should be used for in-situ checking of rinse water conductivity. ii. For partial run CPUs, the rinse water-specific conductivity should be <2.0 $\mu\text{S}/\text{cm}$ for Drum-type units.
3	CPU common inlet header manual valve	Open
4	Selected CPU inlet & primary outlet manual valves	Open
5	Check status of following pneumatic valves: CPU Air vent, Air inlet, Resin inlet, Resin outlet, DM water inlet, Raise bed, Nozzle flush, Drain, Depressurization, Rinse etc.	Close
6	Elevate system pressure by pressurizing valve	Open (once CPU vessel pressure reaches to CPU inlet pressure, close pressurization valve)
7	CPU service inlet pneumatic valve	Open
8	CPU service outlet secondary	Open

	manual valve	
9	CPU service outlet pneumatic valve	Open
	Steps-1 to 9 should to be repeated for taking second CPU in to service	
9	CPU auto bypass valve position	Bypass valve 50% closed when one CPU is in-service & 100% closed when both CPU are in-service. Total condensate flow and total flow of individual CPU to be matched (when both CPUs are in service). Note: <i>To avoid startup loading on both CPU at a time, secondary manual outlet valve of comparatively fresh CPU may be kept closed during startup period.</i> <i>i. As one CPU is designed to cater to 50% TMCR condensate flow, the condensate flow through CPU should be kept under close observation during unit startup period.</i> <i>iii. The second CPU should be taken into service after start of turbine rolling or as per requirement.</i>
	IMPORTANT NOTES: i. After service of CPU, important parameters like flow, DP-vessel & trap, conductivity, ACC, sodium, etc are to be kept under close observation. ii. If required, adjust manual inlet valves of CPU to get equal flow through all the service-in CPUs.	

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ANNEXURE-6

PROTOCOL -CPU IN SERVICE PARAMETERS

DATE & TIME:

UNIT:

CPU:

CPU PARAMETERS

S.NO	CPU Rinse Parameters	Value	Remarks
1	pH		
2	Sp. Conductivity		
3	Chloride(ppb)		
4	Silica(ppb)		

Auto Bypass Valve position:

Manual Bypass Valve Position:

Resin Inlet/Outlet Valves Position:

CPU Flow:

After Service parameters ACC/Sp.cond:

Any other: Annexure-5 to be followed.

Shift Chemist

Operation Executive

MOM on recommendations of Quarterly Chemistry KT meet:

Dated : 03.11.2023

S.N.	Agenda Points & Action to be taken	Responsibility	Target Date
1	Anion, Cation and mixed resins should be rinsed to conductivity as specified in OGN/OPS/CHEM/020.	Site Chemistry	Immediate effect
2	Already regenerated mixed resin must be rinsed in CRU/CRT/MRSV to conductivity <0.10 uS/cm before transferring to CPU service vessel.	Site Chemistry	Immediate effect
3	Mixed resin must be mixed and rinsed in CPU service vessel to conductivity <0.10 uS/cm (online or offline) before taking into service.	Site Chemistry	Immediate effect
4	Stations facing the issue of low sample pressure during rinse of CPU service vessel, should use portable conductivity meter (low level detection) for checking the rinse water specific conductivity of CPU service vessel before taking the vessel into service.	Site Chemistry	Immediate effect
5	After the service of CPU, important parameters like conductivity, ACC, Sodium, flow, DP-vessel & trap etc. of Service-in CPU are to be kept under close observation.	Site Chemistry & Site Operation	Immediate effect

6	A duly signed protocol by Site operation/Chemistry executive is to be maintained for taking CPU SV into service and isolation of CPU SV. The same is to be included in revised OGN/OPS/CHEM/020.	i.COS-Chem: For including in revised OGN. ii.Site Chemistry & Operationmaintaining of joint protocol.	ii.Immediate effect and to be continued.
7	CPU regeneration and Service Vessel area activity log sheet is to be maintained strictly by concerned chemistry executives as per OGN/OPS/CHEM/020 and reviewed by Shift & main plant chemists, at least once/shift.	Site Chemistry	Immediate effect
8	CPU regeneration and operation should be monitored by the Chemist, looking after the main plant chemistry area.	Site-In-Charge- CPU system	Immediate effect
9	SOP of CPU regeneration process and SV operation is to be maintained in local language.	Site Chemistry	Immediate effect

10	Any Deviations in Cycle chemistry parameters as per action level of immediate shut-down of the Unit defined in OGN/OD should be communicated immediately in the Operation Communication register.	Site Chemistry	Immediate effect
11	Notifications is to be raised for all defects of online instrumentations of CPU/ Main plant SWAS and the same is to be discussed in DPM.	Site Chemistry	Immediate effect
12	P&I Checking CPU system is to be done once/year and record is to be maintained.	Site Chemistry, site C&I.	As per frequency mentioned.

13	At Mouda CPU system, interlock is existing for start of regeneration process only after completion of resin transfer process. The same interlock should be ensured for all other NTPC stations having CPU systems.	All NTPC stations & JVs-site C&I	
14	Trend of all-important parameters like all flow, level, conductivity, concentration analyser, blower & pumps pressure etc. should be available in CPU OWS.	Site C&I	
15	Cross checking of CPU SV resin trap DP transmitter, CPU SV header DP, ACC meter is to be done and notification is to be raised immediately.	Site Chemistry to raise notifications for further action by site C&I.	Immediate effect
16	Trip logic for low power water line flow transmitter should be incorporated in stations where it is not there. Others should ensure the healthiness of flow transmitters.	Site Chemistry in consultation with site C&I.	Immediate effect
17	Line executives at all stations having CPU system should be technically competent to operate and regenerate the CPU system. In case line executives do not have sufficient exposure/experience in operating the CPU system, peer site visit to Supercritical stations for hand-holding purpose is to be done for enhanced competencybuilding. CC-OS to facilitate the same, if required.	Site Chemistry HOD. COS-Chemistry to facilitate, if required.	

18	Regular training of CPU operators is to be ensured by the main plant chemist.	Site Chemistry HOD	Immediate effect
19	Chemistry Shift General to be utilized for learning of CPU regeneration and operation, LMI development, preparation of location wise checklist/supporting documents/SOPs etc for process improvement.	Chemistry HOD at site	Immediate effect
20	Regular interaction with site Operation in Shift General to discuss about main plant chemistry issues and action to be taken in case of any chemistry parameters deviations observed, any issues in CPU operation etc. Site Operation should have clear-cut understanding of the Chemistry parameter limits and action levels in case of excursions in parameters as per OD-OPS- CHEM-001	Heads of Site Chemistry and Operation	Immediate effect
21	Trends of all important CPU process performance parameters should be available in DCS/PI and the same is to be incorporated if not available.	Site C&I	Immediate effect
22	CPU SV mimic with CPU SV in/out and control should be available in unit OWS.	Site C&I	Immediate effect

23	i.During normal operation of units, site Chemistry to continuously monitor, cross-check and match the design condensate flow with the individual service vessel flow (for 100% condensate flow) to check any bypass Valve opening in case of SC units. Any discrepancies should be recorded for corrective actions. ii.The reason for high CPU flow, if any, especially in supercritical units, should be addressed as a priority.	i.Site Chemistry of all Supercritical Units. ii.Site Chemistry of all units with CPU.	Immediate effect
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